Adaptive Transfer Learning

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Abstract

Transfer learning has emerged as a dominant paradigm in machine learning, particularly effective when limited labeled data are available in target domains. While standard transfer learning (STL) methods leverage parameter initialization for fine-tuning downstream models, their reliance on weight transfer limits broader applications and optimization capabilities. This work introduces an **Adaptive Transfer Learning (ATL)** framework to overcome these limitations by integrating Bayesian-Tuning(B-Tuning) with advanced kernel learning techniques. Leveraging Gaussian mixture models, adaptive feature transfer, and expressive kernels such as Radial Basis Function (RBF) and deep kernels, ATL enhances feature transferability, interpretability, and overall performance. Experimental results on diverse computer vision and natural language processing datasets validate the effectiveness of ATL, showcasing its ability to outperform STL and other baselines. This work establishes a pathway for more robust, scalable, and efficient transfer learning methodologies.

1. Introduction

The ability to generalize knowledge from one task to another — such as learning to play the accordion and then applying that knowledge to play the piano — has traditionally distinguished biological intelligence from machine intelligence.

Transfer learning has become a dominant learning paradigm in recent years, especially when there is little available labeled data in the target domain(Zhuang et al., 2020). In addition, more and more large foundation models emerged in recent years, which were trained on the massive volume of data. The learned model parameter can be initialized in a small downstream task-specific model to enhance training efficiency. There is no doubt that this standard transfer learning (STL) approach has had great success. However, reliance on parameter initialization is still a limited way of transferring knowledge because as long as we have a good enough optimizer, we can finally reach global optima.

Despite its success, the standard transfer learning (STL) approach, which relies primarily on parameter initialization, remains limited. As long as optimization algorithms continue to improve, the benefits of parameter initialization diminish, providing only a narrow window for effective knowledge transfer. In addition, parameter transfer only limits in the same architecture models. Furthermore, much of the existing research in transfer learning is concentrated on parameter space transfer, overlooking the potential advantages of a feature space perspective.

To address these challenges, our work introduces an **Adaptive Transfer Learning (ATL)** framework that combines B-Tuning with advanced kernel learning methods. By shifting the focus from parameter transfer to feature transfer, the framework enables more flexible, interpretable, and efficient knowledge transfer across diverse architectures and domains. The approach integrates Gaussian mixture models, Adaptive Feature Transfer (AFT), and expressive kernels, such as Radial Basis Function (RBF) kernels and deep kernels, to optimize feature alignment and transferability.

This research explores the potential of ATL across both computer vision and natural language processing tasks, demonstrating substantial improvements over baseline methods, including STL, knowledge distillation, and AFT. By addressing the inherent limitations of STL and advancing feature-based transfer, this work contributes to the development of robust methodologies for efficient and scalable transfer learning.

2. Related Work

Standard Transfer Learning. Standard transfer learning typically begins by loading pre-trained weights θ into downstream models with the same architecture, followed by fine-tuning on the specific downstream tasks.

Informative Priors over Weights. Shwartz-Ziv et al.

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(2022) argued that standard initialization carries limited information about source tasks. To address this, they proposed a Bayesian transfer learning approach that learns highly informative posteriors from the source domain to serve as priors for downstream tasks. Alternatively, the method modifies the entire loss surface for downstream tasks, enabling more efficient adaptation.

Informative Priors over Features. Traditional transfer learning methods (including the above) focus on weight-based transfer, limiting them to upstream models with the same architecture. In contrast, Hinton (2015) introduced knowledge distillation (KD), which distills the knowledge from an ensemble of models into a single model, enabling cross-architecture knowledge transfer. Feature-based KD further facilitates transfer by regularizing features instead of logits. The objective function for feature-based KD is typically defined as $R_{\mathrm{KD}}(\theta) = \min_V \mathbb{E}_{\Phi,\Psi}\left[|V\Phi - \Psi|^2\right]$, where V is affine transformation matrix to transfer downstream feature space to upstream feature space.

Building on this, You et al. (2022) proposed Bayesian-tuning (B-tuning), a method capable of transferring knowledge from multiple upstream models, outperforming traditional KD. B-tuning leverages LogME (You et al., 2021) to rank pre-trained models and employs a Gaussian mixture approach for model selection. More recently, Qiu et al. (2024) introduced Adaptive Feature Transfer (AFT), which selectively transfers the most relevant pre-trained features for downstream tasks instead of indiscriminately compressing all features. AFT optimizes $R(\theta) = \min_\mu \mathbb{E}_{\Phi,\Psi}\left[|\Phi - \mu\Psi|^2\right]$, focusing on downstream feature spaces in contrast to KD, which operates on upstream feature spaces. This selective approach has demonstrated significant performance gains.

Differences Between Our Approach and Prior Work. Our work builds upon AFT by further exploring the use of expressive kernels and combining B-tuning with AFT to enhance feature transfer effectiveness.

3. Methodology

3.1. Kernel Learning

In AFT (Qiu et al., 2024), authors mention while minimizing the ℓ_2 distance between features is conceptually straightforward, it also introduces statistical and optimization challenges: 1) the optimal variational parameter μ changes every time we update θ in $R(\theta)$; 2) Φ and Ψ are empirical feature distribution over downstream small dataset, which may cause overfitting when we minimize $R(\theta)$.

In order to simplify the optimization problem for the affine transformation matrix μ and overcome the overfitting problem, the authors adopt kernel formulation. Specifically, they use the linear kernel to improve the objective because the behavior of a linear kernel $f(\cdot) = w^\top \phi(\cdot)$ is completely characterized by its kernel $k_\Phi(\mathbf{x}, \mathbf{x}') = \phi(\mathbf{x})^\top \phi(\mathbf{x}')$. Thus, the ℓ_2 distance between features can be replaced with a distance between their kernel functions (feature space distance).

$$
\begin{aligned}
& R(\theta) = \min_\mu \mathbb{E}_{\Phi,\Psi}\left[\|\Phi - \mu\Psi\|^2\right] \\
\Rightarrow \quad & R_{\mathrm{AFT}}(\theta) = \min_\mu \sqrt{\mathbb{E}\left[\left(k_\Phi(X, X') - k_{\mu\Psi}(X, X')\right)^2\right]}
\end{aligned}
$$

However, the RBF kernel $k_{\mathrm{RBF}}(\mathbf{x}, \mathbf{x}') = \exp\left(-\|\mathbf{x} - \mathbf{x}'\|^2/(2\ell^2)\right)$ encodes the inductive bias function values closer together in the input space, in the Euclidean sense, is more correlated. The complexity of functions in the input space is determined by interpretable length-scale hyperparameter ℓ. Our method is to learn this hyperparameter from data by maximizing marginal likelihood of Gaussian process before applying the learned kernel function to calculate feature distance. By learning the kernel hyperparameters through marginal likelihood optimization, the Gaussian Process effectively balances model fit and complexity. Once optimized, the learned kernel is used to compute distances or predict outputs for downstream tasks, where it remains fixed.

$$
\begin{aligned}
\log p(\mathbf{y} \mid \boldsymbol{\gamma}, \mathbf{X}) \propto & -(\mathbf{y} - \boldsymbol{\mu}_{\mathbf{X}})^\top (K_{\boldsymbol{\gamma}} + \sigma^2 I)^{-1} (\mathbf{y} - \boldsymbol{\mu}_{\mathbf{X}}) \\
& - \log|K_{\boldsymbol{\gamma}} + \sigma^2 I|,
\end{aligned}
$$

In addition, in order to learn a more expressive kernel, we can further substitute the RBF kernel with a deep kernel(Wilson et al., 2016), which combines the structural properties of deep learning architectures with non-parametric flexibility of kernel methods. It formulates as $k_\Phi(x, x') = \phi(x)^\top \phi(x')$, where $\phi(x)$ is deep neural net feature extractor.

3.2. A Synergistic Approach: Combining B-Tuning and AFT

When transferring knowledge from multiple pretrained models that capture complementary information, Adaptive Feature Transfer (AFT) and B-Tuning adopt fundamentally different approaches:

The core idea behind AFT is to provide a prior for the downstream model by concatenating all the features from the last layers of the pretrained models. This approach ensures that the downstream task has access to a rich pool of features, regardless of the dimensionality of pre-trained model since only features matter. AFT focuses on identifying and leveraging specific features that are most relevant to the downstream task, effectively filtering out irrelevant information and enabling efficient transfer.

In contrast, B-Tuning utilizes a probabilistic framework by associating each pretrained model with a Gaussian distribution. These distributions represent the uncertainty and variability inherent in the feature representations of the pretrained models. B-Tuning then employs a Gaussian mixture model to combine and weigh the contributions of each pretrained model. This probabilistic weighting ensures that the final representation reflects the complementary strengths of the individual models while managing uncertainty in a principled manner.

Thus we propose an approach to integrate these two algorithms. We keep regularizing the downstream features as AFT did but allocate different learning rates for features from different models (B-tuning) equally adapt features (AFT) according to LogME (Maximum Evidence) value $\{\mathcal{L}_k\}_{k=1}^K$ as a mixture of Gaussian $\Psi = \sum_{k=1}^K \pi_k \mu_k \psi_k$, where $\pi_k = \frac{\exp(\mathcal{L}_k/t)}{\sum_{j=1}^K \exp(\mathcal{L}_j/t)}$, where t is a temperature hyperparameter, and μ_k is transformation matrix from upstream feature space to downstream feature space.

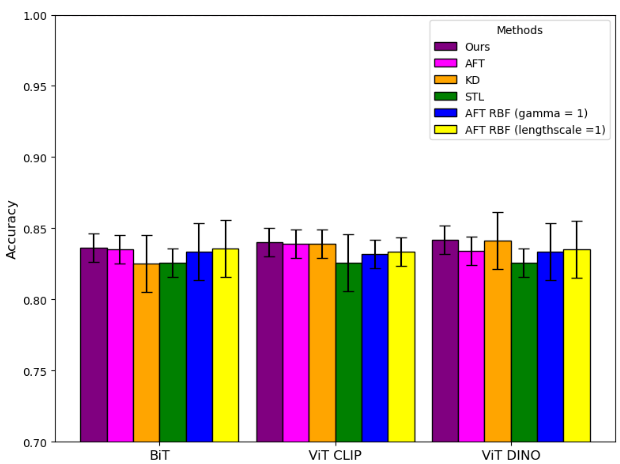


Figure 1. Comparison of Test Accuracy Across Methods and Upstream Models (Downstream Model: ResNet-50, Dataset: CIFAR100).

4. Experiments

We evaluate the proposed method **Adaptive Transfer Learning (ATL)** across a variety of vision and language datasets. To probe the effectiveness of our method in the practical scenario, we transfer from some of the largest and strongest open-source pre-trained vision and language models, such as ViT-G/14 trained with DINOv2(Oquab et al., 2023) and FlanT5(Longpre et al., 2023). We mainly compared with three baselines: AFT, KD, STL.

Algorithm 1 A Synergistic Adaptive Transfer Learning

Symbols:
Pre-computed pre-trained upstream model features Ψ, downstream model features and feature extractor $\Phi = \phi_\theta(X)$, downstream model $f_\theta = W \circ \phi_\theta$, loss function L, batch size B, learning rates (η_1, η_2), regularization coefficient β.

1: Compute LogME values of each model $\{\mathcal{L}_k\}_{k=1}^K$ and Gaussian mixture coefficient $\pi_k = \frac{\exp(\mathcal{L}_k/t)}{\sum_{j=1}^K \exp(\mathcal{L}_j/t)}$.
2: **for** each mini-batch $X_{\text{batch}} \in \mathbb{R}^{B\times d_{\text{in}}}, Y_{\text{batch}} \in \mathbb{R}^{B\times d_{\text{out}}}, \Psi_{\text{batch}} \in \mathbb{R}^{B\times d_\psi}$ **do**
3: Compute features $\Phi_{\text{batch}} = \phi_\theta(X_{\text{batch}}) \in \mathbb{R}^{B\times d_\phi}$ and outputs $\hat{Y}_{\text{batch}} = \Phi_{\text{batch}} W^\top$
4: Mix pre-trained features $\Psi_{\text{batch}} = \sum_{k=1}^K \pi_k \mu_k \psi_k$
5: Subtract the mini-batch mean from Φ_{batch} and Ψ_{batch} and normalize each row
6: Compute $B \times B$ mini-batch kernels $K^{\Phi}_{\text{batch}} = \Phi_{\text{batch}}\Phi_{\text{batch}}^\top$, $K^{\mu\Psi}_{\text{batch}} = \Psi_{\text{batch}}\Psi_{\text{batch}}^\top$
7: Compute mini-batch loss $\hat{L}(\theta) = L(\theta, Y_{\text{batch}}, \hat{Y}_{\text{batch}})$ and the kernel distance estimate:

$$\hat{\delta}(\theta, \mu) = \frac{1}{B}\left\| K^{\Phi}_{\text{batch}} - K^{\mu\Psi}_{\text{batch}} \right\|_F$$

8: Update θ and μ:

$$\theta \leftarrow \theta - \eta_1 \nabla_\theta \left(\hat{L}(\theta) + \beta\hat{\delta}(\theta, \mu)\right)$$

$$\mu_k \leftarrow \mu_k - \eta_{2,k} \nabla_{\mu_k} \hat{\delta}(\theta, \mu_k)$$

9: **end for**

4.1. Computer Vision Tasks

We conducted experiments on image classification tasks to evaluate our approach. The datasets used include CIFAR-100 (Krizhevsky et al., 2009), Flowers (Nilsback & Zisserman, 2008), and Aircraft (Maji et al., 2013), each comprising approximately 100 classes. For the upstream models, we utilized ViT-G CLIP, ViT-G DINO, and BiT 152x4, while the downstream models were based on ResNet-50.

Figure 1 illustrates that AFT with a linear kernel generally outperforms other methods, including standard transfer learning, knowledge distillation, and AFT with an RBF kernel lacking a coefficient $k_{\text{RBF}}(\mathbf{x}, \mathbf{x}') = \exp\left(-\|\mathbf{x} - \mathbf{x}'\|^2\right)$. However, the RBF kernel with a coefficient $k_{\text{RBF}}(\mathbf{x}, \mathbf{x}') = \exp\left(-\frac{1}{2}\|\mathbf{x} - \mathbf{x}'\|^2\right)$ slightly surpasses the linear kernel AFT in performance. Notably, the length-scale-adjusted RBF AFT achieves approximately a 10% improvement over linear kernel AFT, compared to the improvement gain observed from standard transfer learning (STL).

In Figure 2, we can see contrastive performance gain between different datasets, which further support our method.

In Figure 4, we analyze the impact of RBF kernel coefficients, $\frac{1}{2\ell^2} \in \{0.1, 0.5, 10, 1100\}$, and observe that both BiT and DINO models show optimal performance when the coefficient equals 0.5 if we manually adjust the length-scale parameter. This finding motivates further exploration of the relationship between features and length scale, as well as the interplay between the upstream model architecture and length scale, as demonstrated by CLIP.

These results also inspire the investigation of more expressive kernel functions, such as deep kernels or hybrid approaches that combine linear and RBF kernels, to enhance transferability.

4.2. Natural Language Processing Tasks

On natural language processing (NLP) applications, we explored knowledge transfer from FlanT5-B to FlanT5-S (Longpre et al., 2023) and from FlanT5-B to BERTSmall (Kenton & Toutanova, 2019). We evaluate the knowledge transfer experiments will focus on tasks such as sentiment analysis on the Large Movie Review (IMDB) dataset (Maas et al., 2011) and question answering on BoolQ (Wang et al., 2019). See our NLP experiment results in Figure 3.

4.3. Ablation Studies

We propose a modified version of AFT by incorporating elements of B-Tuning into the framework. While maintaining the core focus on feature-based transfer, this enhanced approach introduces different learning rates for features originating from different models. The rationale behind this modification is to allow the algorithm to prioritize information from models that provide stronger evidence, as indicated by their performance on upstream tasks. By assigning higher learning rates to features from models with greater evidence, the method encourages faster refinement of these features, ultimately saving both data resources and computational time.

This approach shares conceptual similarities with STL in that model information is implicitly encoded through the learning rate. Unlike traditional informative priors used in many transfer learning methods, this modification adjusts the learning process without altering the final optimization objective. It emphasizes efficiency in feature transfer rather than imposing additional complexity on the overall optimization.

The methodology aligns with principles from Bayesian machine learning concerning uncertainty management. In Bayesian frameworks, marginalization is often favored over direct optimization to account for uncertainty. However, analogous to Bayesian linear regression—where marginalizing parameters and optimizing hyperparameters often suffices—our approach demonstrates that marginalizing further in this context does not yield significant benefits. Instead, the combined use of AFT and an STL-inspired learning rate adjustment represents a practical and efficient composition of transfer learning strategies. Rather than seeking an entirely new 'optimal' method, this hybrid approach focuses on transferring knowledge through refined feature learning processes, acknowledging that further methodological changes are unlikely to result in substantial improvements. Furthermore, B-Tuning embodies a fundamentally Bayesian perspective by addressing uncertainty at the level of the model itself. Unlike traditional approaches that rely solely on selecting the highest-evidence or fully optimized model, B-Tuning takes into account a broader set of possibilities. Specifically, it considers the top models, incorporating their collective contributions into the learning process. By integrating information from multiple models rather than committing to a single, potentially suboptimal choice, B-Tuning effectively reduces the risk of overfitting or reliance on a model that might not generalize well to unseen data.

5. Discussion

This experiment results highlight the **Adaptive Transfer Learning** framework's generalizability, as evidenced by its effectiveness across both computer vision and natural language processing tasks, thereby validating its broader applicability. Here, we present an in-depth examination of sensitivity analysis, transfer efficiency, computational overhead, and the inherent limitations associated with AFT models.

5.1. Sensitivity Analysis

As we can see in Figure 4, we compared test accuracy across different kernel lengthscales to show how sensitive the performance of the downstream tasks is to variations in the kernel parameter (γ). The AFT with RBF Kernel approach demonstrates notable variations in accuracy as the length-scale parameter γ is adjusted (from 0.1 to 100), reflecting the importance of properly tuning this parameter for achieving optimal results. Specifically, higher or lower values of γ appear to degrade the performance across all three upstream models—BiT 152x4 (blue), ViT CLIP Giant (green), and ViT DINO Giant (orange).

Our proposed approach consistently demonstrates superior or comparable accuracy to other configurations of the AFT model with an RBF kernel across the three upstream models evaluated in our experiments. This suggests that our method achieves superior performance through kernel learning. Notably, the ViT CLIP Giant model achieves the highest test accuracy with our method, indicating its strong generalization ability when paired with our approach.

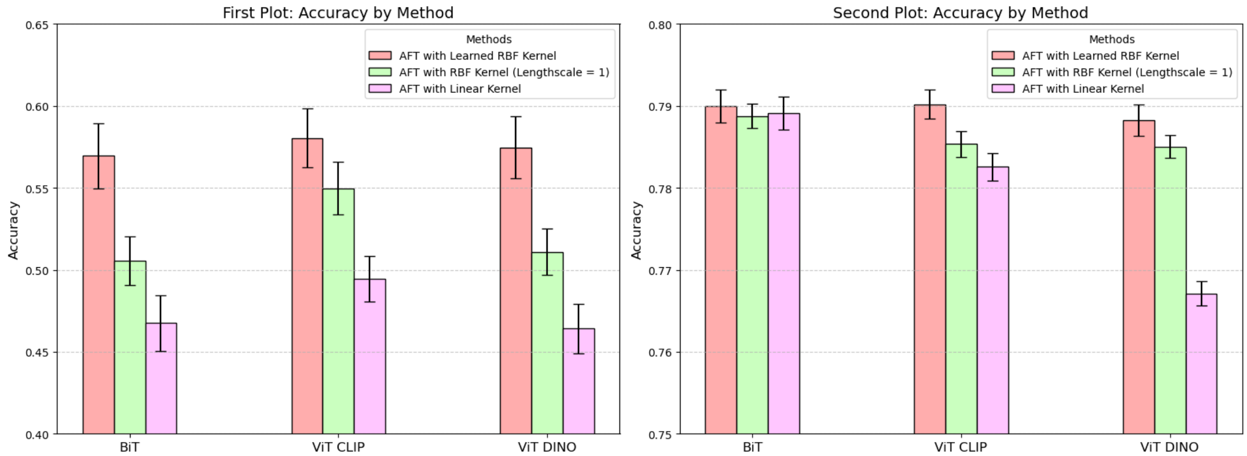


Figure 2. Left: Test Accuracy Comparison of Models Between linear AFT Kernel and RBF kernel (Downstream Model: ResNet-50, Dataset: Aircraft). Right: Test Accuracy Comparison of Models Between linear AFT Kernel and RBF kernel (Downstream Model: ResNet-50, Dataset: Flowers).

Additionally, error bars illustrate the variability across runs, where our method maintains relatively low uncertainty compared to alternative configurations, further emphasizing its stability.

5.2. Transfer Efficiency

Data efficiency refers to the ability of a transfer learning method to effectively utilize pre-trained model information when the available target training data is limited. A data-efficient transfer learning method can extract relevant knowledge and achieve strong performance on downstream tasks, thereby reducing the overall cost and reliance on large datasets. In this part, we compare and analyze the data efficiency of several transfer learning methods and investigate the reasons.

We observed that the performance of transfer learning methods on target tasks varies according to several factors.

Pretrained model Domain The performance of a transfer learning method is significantly influenced by the similarity between the domain of the pre-trained model and that of the downstream target task. From our experiments, we conclude that when the pre-trained model's domain aligns closely with the target one, performance improves significantly: a relatively small amount of data would promise strong performance, thus a high efficiency is promised. However, in many practical scenarios, the upstream model and downstream task are likely to have very different domains, this can reduce the informativeness of transferred knowledge, and irrelevant knowledge may even degrade the downstream model's performance as shown in Figure 1, the same downstream model exhibits varying behavior when provided with knowledge from different upstream models.

Feature-Based Prior To address the challenges posed by domain and architectural mismatches, focusing on feature transfer rather than direct parameter transfer proves advantageous. Features typically have more relevant information regarding the downstream task, enabling a more fine-grained selection of relevant information. By operating on features, transfer learning methods can isolate and utilize the most informative aspects of the upstream model, while discarding irrelevant or harmful knowledge.

Thus, we anticipate that transfer learning methods capable of identifying and leveraging relevant knowledge will exhibit superior data efficiency. KD focuses on feature transfer and is expected to perform well due to its emphasis on features. AFT further enhances data efficiency by adaptively promoting related features while reducing the influence of irrelevant ones and is expected to perform even better by updating the dimension transformation matrix μ. Consequently, we expect AFT to achieve the highest data efficiency, followed by KD, with Standard Transfer Learning (STL) yielding the lowest efficiency.

Our experiments confirm these conclusions. We observed that widely used datasets, such as CIFAR-10 and CIFAR-100, consistently yield strong performances across all three methods, making it challenging to distinguish their relative efficiencies. To better identify performance patterns, we opted to use the Aircraft dataset, which presents more persuasive results. When the upstream model is ResNet101x1 and the downstream model is Mixer-B—a case where the upstream and downstream models differ significantly, AFT clearly demonstrates the highest data efficiency, followed

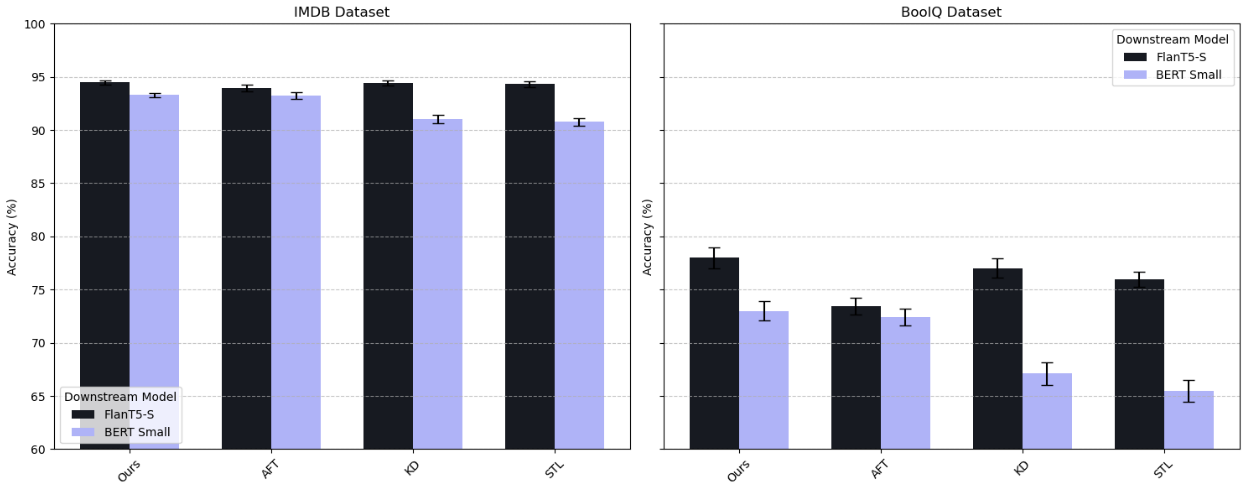


Figure 3. NLP tasks tested across different methods on IMDB and BoolQ dataset. (Downstream model: FlanT5-B)

by KD, with STL performing the least efficiently. In contrast, when both models share similar architectures, such as ResNet101x1 as the upstream and ResNet50x1 as the downstream, the differences in efficiency among AFT, KD, and STL become minimal. This observation further validates our hypothesis.

Additionally, we evaluated data efficiency in multi-source transfer learning scenarios. AFT, which prioritizes feature transfer, demonstrated superior performance. This was evident in experiments involving upstream models BiT 50x1 and BiT 101x1 with the downstream Mixer-B model, as demonstrated in Figure 7. These results consistently confirm that methods capable of selecting and utilizing the most relevant features, such as AFT, achieve the highest data efficiency.

5.3. Computational Overhead

Our experiments confirmed our expectations. While the final performance in terms of accuracy remains comparable across methods, our approach demonstrates a significant improvement in efficiency, reducing the overall training time by approximately 10 percent. This is achieved by dynamically varying the growth rate of features derived from different models, prioritizing the more informative features, and accelerating their optimization. This adjustment not only saves computational resources, but also streamlines the learning process.

Furthermore, the experiments reveal that under conditions of limited data availability, our method yields a noticeable improvement in data efficiency. By focusing on selectively enhancing the most relevant features, the method effectively compensates for the constraints imposed by the smaller dataset. This advantage underscores the robustness of our approach in scenarios where data resources are scarce, enabling better performance without requiring additional data or computational complexity.

5.4. The Inherent Flaws of AFT

As we can see in Figure 6, the performance of the AFT model significantly lags behind both KD and STL. These results reveal that AFT's adaptive fine-tuning process, while theoretically promising, fails to match or surpass standard transfer learning and knowledge distillation approaches on the Aircraft Dataset. Its inability to capture robust representations from the source domain or to adapt well to smaller datasets makes it less practical for real-world scenarios involving limited labeled data. This situation also holds when the dataset is given by CIFAR10 and CIFAR100, in which case all the accuracies are relatively high.

Moreover, the AFT method has a notable shortcoming in that it requires a non-negligible amount of time to extract features, which are often high-dimensional and computationally intensive to process. This limitation becomes particularly pronounced in scenarios involving large datasets where AFT does not consistently promise a significant improvement in performance compared to simpler methods like STL, thus the trade-off between computational cost and performance gains may not always justify its use. This makes AFT less appealing for tasks where computational resources are constrained.

In contrast, our proposed adaptive transfer learning method not only mitigates this shortcoming but also achieves consistent performance improvements across all dataset sizes. This adaptability positions it as a more reliable solution for

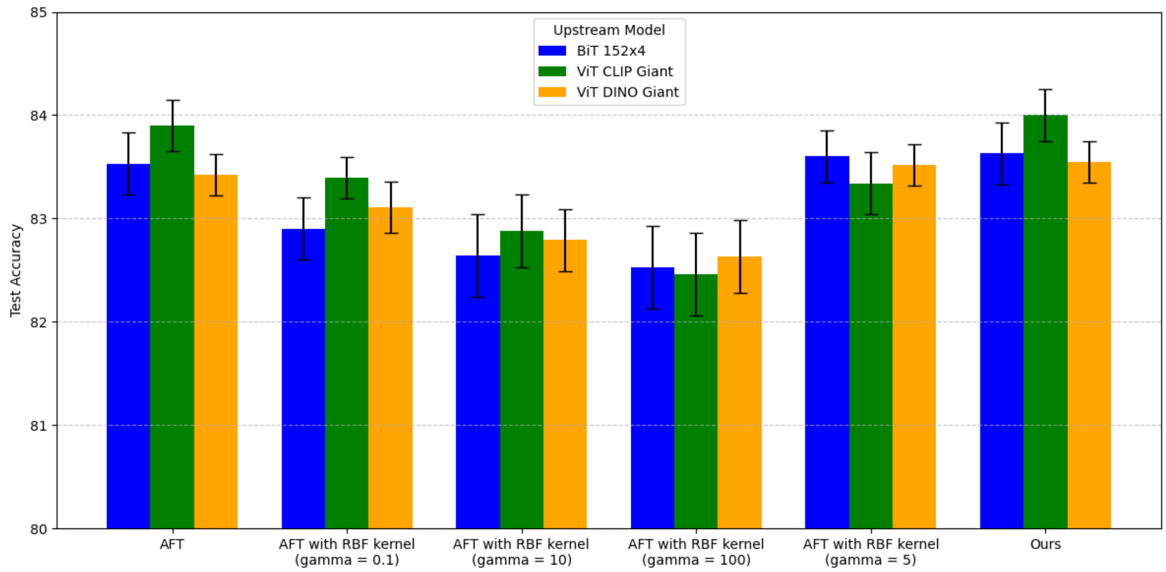
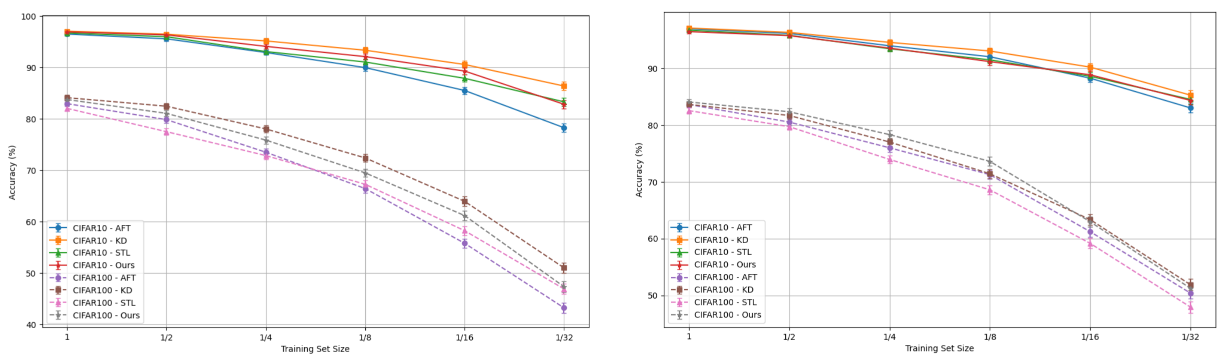


Figure 4. Test Accuracy Comparison of Models Across Different AFT Kernel (Downstream Model: ResNet-50, Dataset CIFAR100).



(a) Upstream: ViT DINO G, Downstream: ResNet50.

(b) Upstream: ViT CLIP Giant, Downstream: ResNet50.

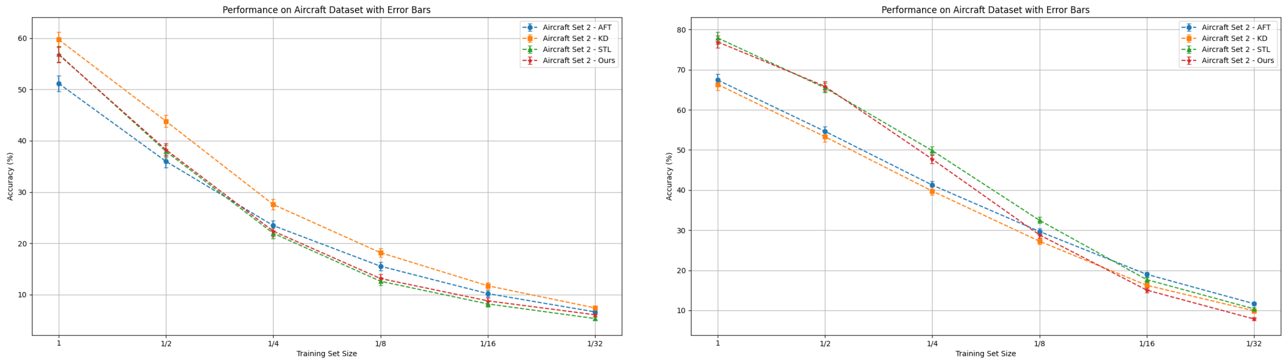
Figure 5. Transfer Efficiency Comparison in CIFAR-10 and CIFAR-100.

transfer learning tasks with constrained resources.

We also acknowledge that AFT may have a potential limitation. While it adaptively adjusts the weights of features based on their relevance to the downstream task, the defined prior is determined using conditional entropy. This metric, although effective, might not always be the most suitable measure for determining feature importance in all scenarios. Moreover, the adaptive adjustment of feature weights essentially aims to achieve an optimal allocation of weights across all features—an optimization process.

From a Bayesian perspective, however, weights should be treated as uncertainties rather than deterministic parameters. Instead of directly optimizing the weight allocation, a more rigorous approach would involve marginalizing over all possible weight configurations. This entails integrating over the entire space of potential weight allocations to account for uncertainties and arrive at a solution that reflects the combined influence of all plausible configurations. Such an approach aligns with Bayesian principles, which emphasize managing uncertainty through probabilistic inference rather than deterministic optimization.

Adopting a Bayesian framework in this context could potentially enhance AFT's performance. By treating weights as random variables and incorporating prior distributions, we could achieve a more robust feature selection process that better reflects the inherent variability in the data. This adjustment could lead to improved generalization, particularly in tasks with limited data or high uncertainty, and represents a promising avenue for future work on refining AFT



(a) Upstream Model: ViT CLIP G, Downstream Model: ResNet 50 (b) Upstream Model: BiT 101x1, Downstream Model: BiT 50x1

Figure 6. Transfer Efficiency Comparison in Aircraft Dataset.

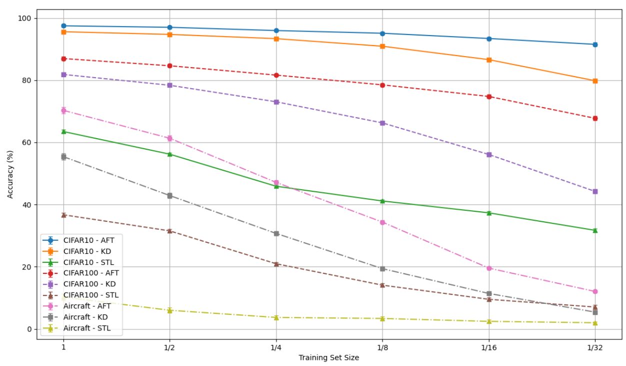


Figure 7. Transfer Efficiency Comparison in CIFAR10, CIFAR100, Aircraft Dataset (Upstream Models: BiT 50x1, BiT 101x1, Downstream Model: B Mixer).

6. Conclusion

This novel **Adaptive Transfer Learning** is a combination of B-Tuning, AFT, and kernel learning methods and represents an innovative contribution to transfer learning. The framework is closely aligned with the principles of Bayesian machine learning by introducing an informative prior tailored specifically for the downstream model. The framework achieves significant improvements in computer vision and natural language processing across multiple datasets, with clear performance gains over baselines.

Despite its notable advancements, the framework's sensitivity to kernel parameter tuning and increased computational overhead highlight areas for further refinement. Future research should focus on optimizing kernel hyperparameters and exploring lightweight implementations to balance computational efficiency with performance gains. Additionally, expanding the framework to include multi-modal and cross-domain transfer learning scenarios will further validate its robustness and applicability.

Overall, this work lays a solid foundation for adaptive transfer learning, offering promising directions for improving knowledge transfer methods in machine learning while addressing real-world challenges in data efficiency and model interoperability.

References

Hinton, G. Distilling the knowledge in a neural network. *arXiv preprint arXiv:1503.02531*, 2015.

Kenton, J. D. M.-W. C. and Toutanova, L. K. Bert: Pre-training of deep bidirectional transformers for language understanding. In *Proceedings of naacL-HLT*, volume 1, pp. 2. Minneapolis, Minnesota, 2019.

Krizhevsky, A., Hinton, G., et al. Learning multiple layers of features from tiny images. 2009.

Longpre, S., Lu, Y., Hou, L., et al. Scaling instruction-finetuned language models. *arXiv preprint arXiv:2210.11416*, 2023.

Maas, A., Daly, R. E., Pham, P. T., Huang, D., Ng, A. Y., and Potts, C. Learning word vectors for sentiment analysis. In *Proceedings of the 49th annual meeting of the association for computational linguistics: Human language technologies*, pp. 142–150, 2011.

Maji, S., Rahtu, E., Kannala, J., Blaschko, M., and Vedaldi, A. Fine-grained visual classification of aircraft. *arXiv preprint arXiv:1306.5151*, 2013.

Nilsback, M.-E. and Zisserman, A. Automated flower classification over a large number of classes. In *2008 Sixth Indian conference on computer vision, graphics & image processing*, pp. 722–729. IEEE, 2008.

Oquab, M., Darcet, T., Moutakanni, T., Vo, H., Szafraniec, M., Khalidov, V., Fernandez, P., Haziza, D., Massa, F., El-Nouby, A., et al. Dinov2: Learning robust visual features without supervision. *arXiv preprint arXiv:2304.07193*, 2023.

Qiu, S., Han, B., Maddix, D. C., Zhang, S., Wang, Y., and Wilson, A. G. Transferring knowledge from large foundation models to small downstream models. *arXiv preprint arXiv:2406.07337*, 2024.

Shwartz-Ziv, R., Goldblum, M., Souri, H., Kapoor, S., Zhu, C., LeCun, Y., and Wilson, A. G. Pre-train your loss: Easy bayesian transfer learning with informative priors. *Advances in Neural Information Processing Systems*, 35: 27706–27715, 2022.

Wang, A., Pruksachatkun, Y., Nangia, N., Singh, A., Michael, J., Hill, F., Levy, O., and Bowman, S. Superglue: A stickier benchmark for general-purpose language understanding systems. *Advances in neural information processing systems*, 32, 2019.

Wilson, A. G., Hu, Z., Salakhutdinov, R., and Xing, E. P. Deep kernel learning. In *Artificial intelligence and statistics*, pp. 370–378. PMLR, 2016.

You, K., Liu, Y., Wang, J., and Long, M. Logme: Practical assessment of pre-trained models for transfer learning. In *International Conference on Machine Learning*, pp. 12133–12143. PMLR, 2021.

You, K., Liu, Y., Zhang, Z., Wang, J., Jordan, M. I., and Long, M. Ranking and tuning pre-trained models: A new paradigm for exploiting model hubs. *Journal of Machine Learning Research*, 23(209):1–47, 2022.

Zhuang, F., Qi, Z., Duan, K., Xi, D., Zhu, Y., Zhu, H., Xiong, H., and He, Q. A comprehensive survey on transfer learning. *Proceedings of the IEEE*, 109(1):43–76, 2020.